Greedy Wins, Rankings Hold: Measuring Decoding-Strategy Sensitivity of Open LLMs Across Four Task Domains

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Abstract

LLM benchmark scores are reported under many different and often unstated decoding settings: greedy for leaderboards, $T{\approx}0.7$ for chat, and sampled pass@1 for code. If scores, or worse, model *rankings*, depend on the decoder, comparisons across papers are fragile. Prior studies of decoding usually look at one task family at a time and do not measure ranking stability. We run a single controlled grid: 11 open-weight instruction-tuned LLMs (7 families, 0.5B–9B parameters), four domains (factual QA, math reasoning, code synthesis, creative writing), and six decoding strategies (greedy, $T{=}0.7$, $T{=}1.0$, top-p 0.9, top-k 40, min-p 0.1), with three seeds per sampled strategy, for 132k generations in total. In all three verifiable domains, every sampler lowers accuracy on average relative to greedy decoding. Untruncated $T{=}1.0$ costs 4.3 points on TRIVIAQA, 5.9 on GSM8K, and 6.7 on HUMANEVAL. Top-p and min-p recover about half of this loss. Under an LLM-judge proxy, sampling does not improve creative quality (at most $+0.7\%$), and $T{=}1.0$ lowers it by 7.9%. Model rankings are robust: Kendall τ against the greedy ranking is at least 0.85 in every verifiable domain, and τ falls to 0.71 only for creative writing at $T{=}1.0$. Individual models can still move by 8–15 points. Code is the most seed-sensitive domain: 27% of problems flip between pass and fail across seeds. We recommend greedy or min-p/top-p decoding for evaluation, explicit reporting of decoding settings, and averaging over at least three seeds when sampling.

1 Introduction

A benchmark number for a large language model (LLM) is the product of the model *and* the decoder that turns its next-token distributions into text. Decoding settings are rarely held fixed across reports. Leaderboards use greedy decoding, chat deployments use temperatures near 0.7, and code papers report pass@1 sampled at temperatures between 0.2 and 0.8 [Chen et al., 2021]. Many papers do not state the setting at all. If a domain's scores depend strongly on the decoder, cross-paper comparisons in that domain are unreliable. If its *model rankings* depend on the decoder, even the qualitative conclusion "model A beats model B" is in question.

Why does this matter now? Practitioners routinely choose between greedy decoding and a range of samplers (temperature, top-k, nucleus/top-p [Holtzman et al., 2020], and the more recent min-p [Nguyen et al., 2024]) to trade accuracy for diversity. They need to know which tasks tolerate sampling and which sampler loses the least. Evaluators need to know whether a ranking computed under one decoder carries over to another.

What is missing? Prior work mostly studies decoding one task family at a time. Nucleus sampling was proposed for open-ended generation [Holtzman et al., 2020], temperature was studied against pass@k for code [Chen et al., 2021], and temperature was found to have little effect on multiple-choice problem solving [Renze and Guven, 2024]. The broadest recent examination [Shi et al.,

39th Conference on Neural Information Processing Systems (NeurIPS 2025).

2024] covers many tasks but few models and does not analyze ranking stability. To our knowledge, no single controlled grid of models $\times$ domains $\times$ samplers $\times$ seeds measures score sensitivity, ranking stability, and seed variance side by side.

Our approach. We run 11 open-weight instruction-tuned LLMs from 7 families on four domains: factual QA (TRIVIAQA), multi-step math reasoning (GSM8K), code synthesis (HUMANEVAL), and creative story writing (ROCSTORIES prompts scored by an LLM judge). Each model is decoded with six strategies, and every sampled strategy is run with three seeds. This gives 132,064 generations, all served locally with VLLM [Kwon et al., 2023]. We pre-register five quantitative hypotheses (H1–H5 in section 3) and test each one with paired statistics and item-bootstrap confidence intervals.

What do we find? Sampling never helps on average in a verifiable domain. Pure $T{=}1.0$ sampling costs 4.3 points on factual QA (an 11.0% relative drop), 5.9 on GSM8K, and 6.7 on HUMANEVAL. top-k 40 behaves almost exactly like $T{=}1.0$. Truncated samplers are much safer: top-p 0.9 recovers 50% of the GSM8K loss and min-p 0.1 recovers 57%. Factual QA is *not* decoding-invariant; its relative loss matches that of reasoning. Sampling does not raise LLM-judged creative quality: changes range from $+0.0\%$ to $+0.7\%$ and are not significant, and $T{=}1.0$ costs 7.9%. Model rankings are reassuringly stable. Kendall τ against the greedy ranking is 0.93 on factual QA, 0.85–0.89 on reasoning, and 0.93–1.00 on code. The only τ near 0.7 comes from small models degenerating on creative writing at $T{=}1.0$. Code is the noisiest domain across seeds, with a seed standard deviation of 2.13 points and 26.7% of problems flipping outcome. About half of that gap disappears once we control for benchmark size.

Our contributions are:

- We conduct a controlled 11-model $\times$ 4-domain $\times$ 6-strategy $\times$ 3-seed study of decoding sensitivity, with all samplers implemented identically in one serving stack.
- We test five quantitative hypotheses about domain-specific sensitivity and refute or qualify four of them. Factual QA is not robust to sampling (H1), the reasoning gap is strongly model-dependent (H2), sampling does not improve judged creativity (H3), and reasoning rankings are less stable than factual ones but far from $\tau < 0.7$ (H4).
- We measure ranking stability with item-bootstrap Kendall τ and seed variability with a matched-n control. We show that rankings shift mainly when models are tightly packed or when small models degenerate.
- We derive practical recommendations: use greedy or min-p/top-p for evaluation, never raw $T{=}1.0$ or top-k 40, and average at least three seeds when sampling, especially for code.

The rest of the paper is organized as follows. Section 2 reviews related work, section 3 describes the experimental design, section 4 presents results, section 5 discusses implications and limitations, and section 6 concludes.

2 Related Work

Truncated sampling for open-ended generation. Holtzman et al. [2020] showed that greedy and beam search produce bland, repetitive text, while pure ancestral sampling falls into the unreliable tail of the distribution. They proposed nucleus (top-p) sampling as a compromise. Top-k sampling [Fan et al., 2018] truncates to a fixed number of candidates. Min-p sampling [Nguyen et al., 2024] keeps tokens whose probability is at least a fraction of the top token's probability, so the cut-off adapts to model confidence. These methods were motivated mainly by open-ended text quality. We evaluate all of them on verifiable tasks as well, and we find that the adaptive samplers (top-p, min-p) are also the least harmful there.

Decoding for reasoning and code. For code, Chen et al. [2021] showed that low temperatures maximize pass@1 and higher temperatures help pass@k for large k. For multi-step reasoning, sampling is used deliberately in self-consistency [Wang et al., 2023], which aggregates many sampled chains. Renze and Guven [2024] reported that temperature in the range 0–1 has little effect on multiple-choice problem solving. Unlike these single-domain studies, we compare domains under the same models and samplers. This lets us show that free-form factual QA is about as sensitive in relative terms as math reasoning.

Broad examinations of decoding. Shi et al. [2024] examine many decoding methods across tasks for modern LLMs and find that the best method depends on the task and the model. Our study is complementary: it uses more models (11 from 7 families), repeated seeds, and two analyses they do not report, namely *ranking stability* measured by Kendall τ with item-bootstrap confidence intervals, and *seed variance* by domain.

Evaluation robustness and LLM judges. Benchmarks are known to be sensitive to evaluation choices such as prompt format. Decoding is one such choice, and it is often unreported. For open-ended outputs, LLM-as-a-judge [Zheng et al., 2023] has become a common proxy for human ratings, with known biases toward fluent text. We use a held-out LLM judge for creative writing and report diversity metrics alongside it. Our creative-writing conclusions are therefore stated relative to this proxy.

3 Experimental Design

Our main question is: **how much do task scores and model rankings change with the decoding strategy, and does this differ by task domain?** We fix models, prompts, and data, and vary only the decoder.

3.1 Hypotheses

We test five pre-registered claims:

H1 On factual QA, every strategy stays within 2 points of greedy accuracy.

H2 On math reasoning, T=1.0 is 5–12 points below greedy, and top-p 0.9 recovers most (>50%) of that gap.

H3 Sampling improves judged creative quality by 10–15% (relative) over greedy.

H4 Kendall τ between the greedy ranking and each strategy's ranking is > 0.9 on factual QA and < 0.7 on reasoning.

H5 Seed variance under sampling is largest for code.

3.2 Models

We evaluate 11 instruction-tuned open-weight models from 7 families, spanning 0.5B to 9B parameters: Qwen2.5-{0.5B, 1.5B, 3B}-Instruct (bf16), Qwen2.5-7B-Instruct (AWQ) [Qwen Team, 2024], Qwen3-4B-Instruct-2507 (bf16) [Qwen Team, 2025], Llama-3.1-8B-Instruct (AWQ-INT4) [Llama Team, AI @ Meta, 2024], Gemma-2-9B-it (AWQ-INT4) [Gemma Team, 2024], Mistral-7B-Instruct-v0.3 (AWQ) [Jiang et al., 2023], Falcon3-7B-Instruct (AWQ), Phi-3-mini-4k-instruct (GPTQ 4-bit) [Abdin et al., 2024], and Olmo-3-7B-Instruct (GPTQ w4a16). Seven of the 11 checkpoints are 4-bit quantized [Lin et al., 2024, Frantar et al., 2023]. No API keys were available on our host, so we did not evaluate closed models. Open models also give us exact control over samplers (most closed APIs do not expose top-k or min-p) and over per-request seeds.

3.3 Decoding strategies

Table 1 lists the six strategies. All sampled strategies except T=0.7 use T=1.0, so that differences among them come only from truncation. Each sampled strategy is run with seeds $s \in \{0, 1, 2\}$. The per-request seed is $100000 \cdot s + i$ for item index i, which makes outputs independent of batch composition. Greedy decoding is run once. Each model–task pair therefore has $1 + 5 \times 3 = 16$ runs.

3.4 Tasks, data, and metrics

Table 2 summarizes the four domains. All prompts are zero-shot, use each model's chat template, and are identical across models and strategies.

Factual QA. We sample 400 questions (398 unique) from the TRIVIAQA [Joshi et al., 2017] `rc.nocontext` validation split. The model is asked to reply with only the answer. A response is correct if the normalized first line exactly matches, or contains, any gold alias.

Strategy	Parameters	Runs per model/task
GREEDY	$T = 0$ (argmax)	1
T=0.7	$T = 0.7$, no truncation	3
T=1.0	$T = 1.0$, pure ancestral sampling	3
top-p 0.9	$T = 1.0$, nucleus $p = 0.9$	3
top-k 40	$T = 1.0$, $k = 40$	3
min-p 0.1	$T = 1.0$, $p_{\min} = 0.1$	3

Table 1: The six decoding strategies. All truncated samplers use T=1.0, so differences among them come only from truncation.

Domain	Data	Metric	n items	Max new tokens
Factual QA	TRIVIAQA `rc.nocontext` val.	Normalized alias match (EM or contains)	400	48
Reasoning	GSM8K test, zero-shot CoT	Final-number accuracy	400	640
Code	HUMANEVAL (all)	pass@1, executed unit tests	164	640
Creative	ROCSTORIES first sentences	LLM-judge mean of 4 criteria (1–10)	100	400

Table 2: Task domains, data, and metrics. Each item is generated under 16 runs (greedy plus 5 sampled strategies $\times$ 3 seeds) for each of the 11 models.

Math reasoning. We sample 400 GSM8K [Cobbe et al., 2021] test problems. The prompt asks for step-by-step reasoning that ends with "The answer is N". We extract that number, falling back to the last number in the response, and compare it with the gold answer.

Code synthesis. We use all 164 HUMANEVAL [Chen et al., 2021] problems. We extract the code block, concatenate it with the prompt and unit tests, and execute the result in an isolated subprocess with a 10-second timeout. The metric is pass@1 per sample.

Creative writing. We take the first sentences of 100 ROCSTORIES [Mostafazadeh et al., 2016] test stories and ask for an original story of about 150 words that begins with that sentence. The judge is Qwen3-8B-AWQ, run with greedy decoding in non-thinking mode. It is not one of the evaluated models and does not see the model or strategy. It rates each story from 1 to 10 on creativity, coherence, fluency, and engagement, and we report the mean of the four scores. All 17,600 judgments parsed successfully. We also report distinct-2 [Li et al., 2016], length, and truncation rate.

3.5 Statistical analysis

Let $a_{m,d,g,s}$ be the score of model m on domain d under strategy g and seed s, averaged over items. The model-level score is $\bar{a}_{m,d,g} = \frac{1}{3}\sum_s a_{m,d,g,s}$. For greedy, this is the single run. We define the change relative to greedy as

$$\Delta_{m,d,g} = \bar{a}_{m,d,g} - \bar{a}_{m,d,\text{GREEDY}}. \tag{1}$$

Score changes. We test whether $\Delta_{m,d,g}$ is zero across the 11 models with a Wilcoxon signed-rank test (we also run a t-test) and apply Holm correction over all 20 (domain, strategy) pairs.

Ranking stability. For each domain and strategy, we compute Kendall's τ_b between the model ranking under greedy and the ranking under that strategy. We obtain 95% confidence intervals from 1,000 item-bootstrap resamples, recomputing every model mean in each resample. With 11 models, one swapped pair changes τ by $2/55 \approx 0.036$.

Seed variability. For each (model, strategy) cell, we compute the standard deviation (SD) of run-level scores over the three seeds. We also compute the *item flip rate*, the share of items whose outcome differs across seeds. Domains are compared with paired Wilcoxon tests over the 55 model$\times$strategy cells. HUMANEVAL has fewer items than the other benchmarks, so we add a matched-n check that draws 200 subsamples of 164 items from TRIVIAQA and GSM8K.

Compute. We ran generation on 4$\times$ RTX A6000 GPUs shared with other jobs. Each model received a 5–15 GB slice (max context 2,048, at most 128 concurrent sequences). We produced 132,064 generations (about 28.6M tokens) in about 4.5 hours, and judging took about 45 minutes.

Domain (unit)	GREEDY	T=0.7	T=1.0	top-p 0.9	top-k 40	min-p 0.1
Factual (acc. %)	**51.8**	50.2 (-1.6^*)	47.4 (-4.3^*)	49.0 (-2.8^*)	48.0 (-3.8^*)	50.2 (-1.6^*)
Reasoning (acc. %)	**79.3**	77.2 (-2.0)	73.3 (-5.9^*)	76.3 (-2.9^*)	73.5 (-5.7^*)	76.7 (-2.5)
Code (pass@1 %)	**63.5**	61.2 (-2.3)	56.9 (-6.7^*)	60.6 (-3.0)	56.6 (-6.9^*)	61.0 (-2.5)
Creative (judge, 1–10)	8.30	8.33 ($+0.03$)	7.71 (-0.59^*)	8.30 ($+0.00$)	8.30 ($+0.00$)	**8.35** ($+0.05$)

Table 3: Mean score over 11 models for each domain and strategy. The change relative to greedy is in parentheses. *: Wilcoxon signed-rank test over the 11 models, Holm-adjusted $p < 0.05$ across all 20 tests. The best strategy in each row is in **bold**. Greedy is best in every verifiable domain. T=1.0 and top-k 40 lose the most, and T=0.7, top-p 0.9, and min-p 0.1 lose about half as much.

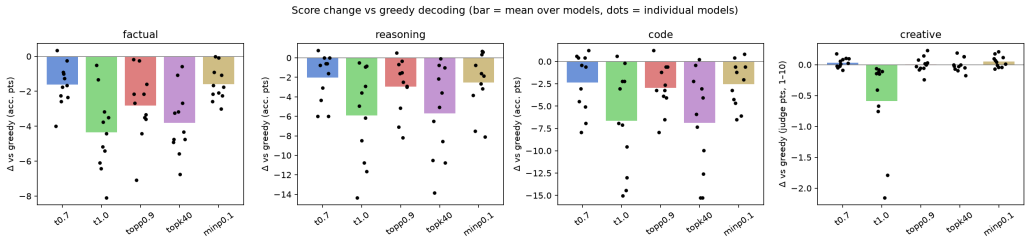


Figure 1: Change in score relative to greedy decoding for each domain and strategy. Bars show the mean over the 11 models, and dots show individual models. In the three verifiable domains, nearly every model loses accuracy under every sampler. Losses under T=1.0 and top-k 40 reach 8–15 points for some models. Creative quality is flat except at T=1.0.

4 Results

We first report how scores change relative to greedy (section 4.1). We then test each hypothesis: reasoning (section 4.2), ranking stability (section 4.3), seed variability (section 4.4), and creative writing (section 4.5).

4.1 Sampling lowers accuracy in every verifiable domain

Table 3 and figure 1 show that greedy decoding gives the highest mean score in all three verifiable domains. No sampler beats it on average. The order of strategies is nearly the same in every domain: greedy $>$ {T=0.7, min-p 0.1, top-p 0.9 } $>$ {top-k 40, T=1.0 }. top-k 40 is almost indistinguishable from untruncated T=1.0. Factual QA drops 3.8 vs. 4.3 points, reasoning 5.7 vs. 5.9, and code 6.9 vs. 6.7. A fixed cut-off of 40 tokens rarely removes anything from the peaked distributions of instruction-tuned models. The adaptive samplers, top-p and min-p, do remove the long tail and lose only 1.6–3.0 points.

H1: factual QA is not decoding-invariant (refuted). All five samplers significantly reduce TRIVIAQA accuracy (Holm-adjusted $p \leq 0.035$). The mean drop ranges from 1.6 points (T=0.7, min-p 0.1) to 4.3 points (T=1.0), or 4.0–11.0% in relative terms. Individual models lose up to 8.1 points. The 2-point bound in H1 holds only for the two mildest samplers, and only on average: under T=0.7 one model still loses 4.0 points. The item-level pattern explains why. Pooled over models, items that greedy answers correctly remain correct in only 87% of T=1.0 samples, while items that greedy gets wrong become correct in only 5% of samples. Sampling mostly turns confident correct answers into plausible wrong ones and rarely rescues a miss. Relative to its lower base accuracy, factual QA loses about as much as reasoning (11.0% vs. 8.6% at T=1.0).

4.2 The reasoning gap is real on average but model-dependent

H2 (mostly supported). The mean GSM8K gap between greedy and T=1.0 is 5.9 points ($p_{\text{Holm}} = 0.020$), inside the hypothesized 5–12-point band. Table 4 shows that the per-model gap ranges from 0.5 to 14.3 points, and only 4 of 11 models fall inside the band. The gap shrinks as greedy accuracy rises. The three strongest reasoners (Qwen3-4B, Qwen2.5-7B, Gemma-2-9B, all above 87% under greedy) lose less than 1 point. Small models lose 12–14 points. Falcon3-7B is an outlier:

Model	Greedy acc.	Gap: greedy $-$ T=1.0	top-p 0.9 recovery	min-p 0.1 recovery
Qwen2.5-0.5B	50.5	14.3	51%	48%
Qwen2.5-1.5B	72.5	11.7	74%	73%
Llama-3.1-8B	85.0	10.8	52%	64%
Falcon3-7B	87.0	8.5	4%	5%
Mistral-7B	55.5	6.2	74%	57%
Phi-3-mini	80.8	4.9	49%	63%
Qwen2.5-3B	84.8	3.6	19%	56%
Olmo-3-7B	88.0	2.9	49%	120%
Qwen2.5-7B	89.2	0.9	155%	173%
Qwen3-4B	**90.8**	0.8	60%	140%
Gemma-2-9B	87.8	**0.5**	$-$33%	$-$50%
Pooled	79.3	5.9	50%	57%

Table 4: Per-model GSM8K gap between greedy and T=1.0, and the share of that gap recovered by truncated sampling: $(\bar{a}_{\text{trunc}} - \bar{a}_{T=1})/(\bar{a}_{\text{greedy}} - \bar{a}_{T=1})$. Only 4 of 11 models fall in the hypothesized 5–12-point band. The gap shrinks as greedy accuracy rises, with Falcon3-7B as a notable exception. Recovery ratios are unstable when the gap is below about 1 point.

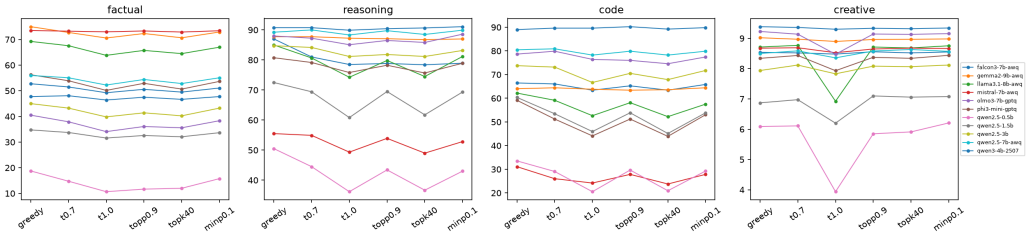


Figure 2: Per-model scores under each strategy. Most lines are nearly parallel, which is why rankings are stable. The visible V shapes at T=1.0 and top-k 40 come from weaker models (Qwen2.5-0.5B/1.5B, Phi-3-mini, Llama-3.1-8B). In the creative panel, Qwen2.5-0.5B and Llama-3.1-8B collapse only at untruncated T=1.0.

it is strong under greedy (87.0%) but loses 8.5 points, and truncation hardly helps it. Pooled over models, top-p 0.9 recovers 50% of the T=1.0 loss and min-p 0.1 recovers 57%. That meets the "$>$50%" threshold only narrowly, so "recovers most" overstates it. Under T=0.7 and min-p 0.1, the reasoning loss is not significant after Holm correction ($p_{\text{Holm}} = 0.13$ and 0.10).

4.3 Model rankings are stable; reasoning is the least stable verifiable domain

H4 (half supported). Table 5 and figure 3 show that factual-QA rankings are stable: $\tau = 0.93$ for every sampler (one swapped pair out of 55), and 85–97% of bootstrap replicates exceed 0.9. As predicted, reasoning is the least stable verifiable domain, with τ between 0.85 and 0.89 and only 10–20% of bootstrap replicates above 0.9. However, τ never comes close to the hypothesized < 0.7. Point estimates are at least 0.85, and even the lower CI bounds are 0.70–0.71. The instability comes from tight clustering rather than from the domain itself. Under greedy, the top five GSM8K models are within 3.75 points of each other (Qwen3-4B 90.8, Qwen2.5-7B 89.2, Olmo-3 88.0, Gemma-2 87.8, Falcon3 87.0), so small sampling effects swap neighbors. Code rankings are the most stable ($\tau = 1.00$ under T=0.7, top-p 0.9, and min-p 0.1), although their CIs are wide because HUMANEVAL is small.

The only domain–strategy pair near $\tau = 0.7$ is *creative writing at* T=1.0 ($\tau = 0.71$ [0.64, 0.81]). Untruncated sampling makes some small models degenerate: Qwen2.5-0.5B loses 35% of its judge score and Llama-3.1-8B loses 21%, while Gemma-2-9B, Mistral-7B, Falcon3-7B, Qwen2.5-7B, and Qwen3-4B lose under 2%. Stable rankings do not imply stable scores, however. Individual models move by up to 8.1 points on factual QA, 14.3 on reasoning, and 15.2 on code (figure 1), so comparing two models under different decoders can reverse the conclusion.

	Kendall τ vs. greedy ranking [95% item-bootstrap CI]					
Domain	T=0.7	T=1.0	top-p 0.9	top-k 40	min-p 0.1	**Mean pairwise τ**
Factual	0.93 [0.89, 1.00]	0.93 [0.88, 1.00]	0.93 [0.89, 1.00]	0.93 [0.89, 1.00]	0.93 [0.89, 1.00]	**0.98**
Reasoning	0.89 [0.71, 0.96]	0.85 [0.71, 0.95]	0.85 [0.70, 0.95]	0.85 [0.71, 0.96]	0.85 [0.70, 0.93]	0.91
Code	1.00 [0.80, 1.00]	0.93 [0.76, 0.96]	1.00 [0.80, 1.00]	0.93 [0.77, 0.99]	1.00 [0.80, 1.00]	0.96
Creative	0.96 [0.93, 1.00]	0.71 [0.64, 0.81]	0.96 [0.92, 1.00]	0.96 [0.89, 1.00]	0.96 [0.93, 1.00]	0.88

Table 5: Ranking stability of the 11 models. The last column averages τ over all 15 pairs of strategies. The most stable domain overall is in **bold**. Every verifiable domain has $\tau \geq 0.85$. The only value near 0.7 is creative writing at T=1.0, where small models degenerate.

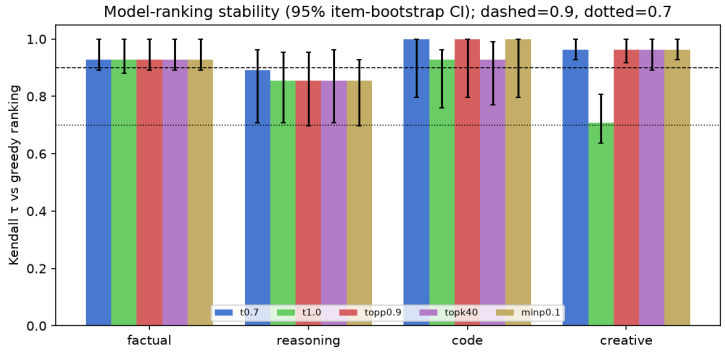


Figure 3: Kendall τ between each strategy's model ranking and the greedy ranking, with 95% item-bootstrap CIs. The dashed line marks 0.9 and the dotted line 0.7. Factual and code rankings stay near or above 0.9. Reasoning sits slightly lower (0.85–0.89) but well above 0.7. Only creative writing at T=1.0 approaches 0.7.

4.4 Code is the most seed-sensitive domain

H5 (supported, with a caveat). Table 6 and figure 4 show that code has the largest run-level seed SD (2.13 points, vs. 1.16 for reasoning and 1.14 for factual) and the highest item flip rate (26.7%). More than a quarter of HUMANEVAL problems change between pass and fail across seeds for the same model and sampler. These differences are significant (paired Wilcoxon over 55 cells, $p < 10^{-5}$; $p = 0.02$ for the flip rate of code vs. reasoning). Part of the raw gap comes from benchmark size. When GSM8K and TRIVIAQA are subsampled to 164 items, their seed SD rises to 1.85 and 1.59, so code's lead over reasoning falls from 0.97 to 0.28 points. Normalized by the binomial standard error, code is still the highest (0.61 vs. 0.58 and 0.49). Creative judge scores average over 100 stories and many tokens, and they barely vary across seeds (SD 0.05 judge points).

4.5 Sampling does not improve judged creative quality

H3 (refuted). Relative to greedy, the overall judge score changes by $+0.5\%$ (T=0.7), $+0.0\%$ (top-p 0.9), $+0.0\%$ (top-k 40), and $+0.7\%$ (min-p 0.1), none of them significant. It falls by 7.9% under T=1.0 ($p_{\text{Holm}} = 0.020$). None of these comes close to the hypothesized $+10$–15%. Table 7 shows why. Sampling does raise the judge's *creativity* sub-score (by 2–4%) and distinct-2 (from 0.963 to 0.973–0.982), but it lowers coherence and fluency by about the same amount. Greedy stories from instruction-tuned models are already fluent and coherent. Untruncated sampling visibly breaks small models: Qwen2.5-0.5B drifts into word salad ("Dory's repertoire of species expert claim proved invaluable…"), and 27% of its T=1.0 stories hit the length cap.

5 Discussion

Does the sampler matter more than the domain? The same order of strategies appears in all three verifiable domains, and the gap between samplers within a domain (for example, 1.6 vs. 4.3 points on factual QA) is as large as the gap between domains. The choice of truncation rule is therefore

Domain	Items	Seed SD (pts)	Seed range (pts)	Item flip rate	SD / binomial SE
Code	164	**2.13**	**4.08**	**26.7%**	**0.61**
Reasoning	400	1.16	2.21	24.2%	0.58
Factual	398	1.14	2.18	14.4%	0.49
Creative (judge pts)	100	0.05	0.09	–	–
Matched $n = 164$ items (200 subsamples)					
Code	164	**2.13**			
Reasoning	164	1.85 [1.61, 2.20]			
Factual	164	1.59 [1.30, 2.05]			

Table 6: Seed variability under sampling, averaged over 11 models $\times$ 5 sampled strategies. The flip rate is the share of items whose outcome differs across the 3 seeds. Code is the most seed-sensitive domain on every measure (paired Wilcoxon over 55 cells, $p < 10^{-5}$; flip rate of code vs. reasoning, $p = 0.02$). With matched n, code's lead over reasoning shrinks to 0.28 points.

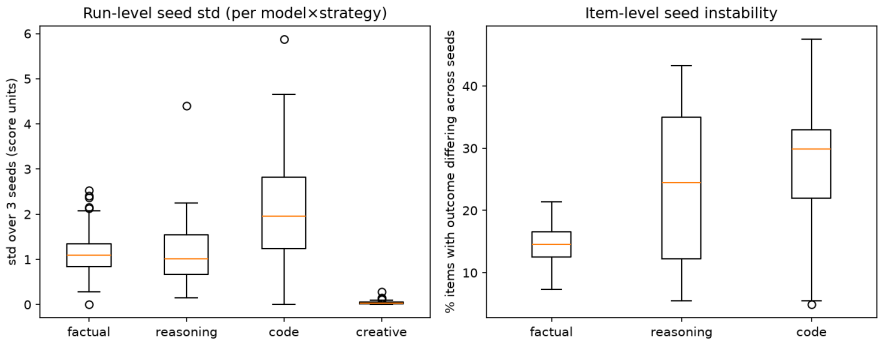


Figure 4: Seed variability by domain. *(Left)*Standard deviation of run-level scores over 3 seeds for each model$\times$strategy cell. *(Right)*Share of items whose outcome differs across seeds. Code has the highest median and the widest spread on both measures. Reasoning flip rates vary widely across models.

at least as important as the task family in predicting the cost of sampling. Adaptive rules (top-p, min-p) are much safer than a fixed top-k. In our grid, min-p 0.1 was the best sampler overall: it stayed closest to greedy on every verifiable task and had the highest creative score.

Why are strong models insensitive? On GSM8K, the models above 87% greedy accuracy lose less than 1 point at T=1.0, and the smallest models lose 12–14 points. A plausible explanation is that stronger models have sharper next-token distributions along their reasoning traces, so even untruncated sampling rarely leaves the high-probability path. This suggests that decoding sensitivity is partly a property of model quality, and that frontier models may be less sensitive than the models we tested. Falcon3-7B is a counterexample: it is accurate under greedy decoding but fragile under sampling.

What drives ranking instability? Ranking changes followed the spacing between models, not the domain. Reasoning rankings moved because the top five models are within 3.75 points of each other. Creative rankings moved at T=1.0 because only some small models degenerated. Reporting τ alone can hide large per-model shifts: models moved by up to 15 points while rankings stayed largely intact. Evaluators should report both.

Practical recommendations.

- Always report the decoding strategy and its parameters. Compare models only under the same decoder.
- Evaluate verifiable tasks with greedy decoding. If diversity is needed, use min-p 0.1, top-p 0.9, or $T \leq 0.7$, never untruncated T=1.0 or top-k 40.
- When sampling, average over at least three seeds, especially on small code benchmarks, where single-seed pass@1 can differ by about 4 points.

	LLM judge (1–10)					Surface statistics		
Strategy	Creativity	Coherence	Fluency	Engagement	Mean	Distinct-2	Words	Truncated
GREEDY	7.24	**8.36**	**9.47**	8.13	8.30	0.963	148	**1.6%**
T=0.7	7.39	8.31	9.44	8.20	8.33	0.973	151	1.8%
T=1.0	7.29	7.37	8.56	7.64	7.71	**0.982**	156	4.5%
top-p 0.9	7.47	8.20	9.34	8.20	8.30	0.977	154	2.8%
top-k 40	**7.53**	8.15	9.31	**8.22**	8.30	0.979	156	2.7%
min-p 0.1	7.39	8.35	9.46	8.21	**8.35**	0.973	153	2.0%

Table 7: Creative writing, averaged over 11 models. Sampling raises creativity (+2 to +4%) and lexical diversity but lowers coherence and fluency, so the overall judge score stays flat or drops. **Bold** marks the best value per column, counting lower truncation as better.

Limitations. *Scale and quantization.* All models have at most 9B parameters, and 7 of 11 are 4-bit quantized. We could not test frontier API models because no API keys were available, and their sensitivity is likely smaller. *Judge proxy.* Creative quality is scored by a single 8B LLM judge, not by humans. LLM judges favor fluent, coherent text [Zheng et al., 2023] and may under-reward novelty. The judge (Qwen3) also belongs to the same family as five of the evaluated Qwen models. H3 is therefore tested only by proxy. *Sample size.* We use 400/398/164/100 items, 3 seeds, and 11 models. With 11 models, τ moves in steps of 0.036, and reasoning CIs span about ± 0.13. *Greedy is a single run.* Batched vLLM inference can make greedy decoding slightly non-deterministic. Our Δ tests average out seed noise for the sampled strategies but not run noise for greedy. *Metrics and prompts.* Alias matching on the first line for TRIVIAQA and number extraction for GSM8K are standard but imperfect. Two TRIVIAQA questions appear twice in our sample. Truncation at the token limit affects T=1.0 slightly more than greedy (4.5% vs. 1.6% of creative outputs). We use one zero-shot prompt per task and do not study interactions with prompt wording. *Literature.* Our survey of related work was not systematic.

Broader impact. This work aims to make LLM evaluation more reproducible. We see no direct negative use. One risk is that the results are read as applying to all LLMs. They describe small and mid-sized open models, and the size of the effect should be checked again for larger systems.

6 Conclusion

We measured how decoding strategy affects the scores and rankings of 11 open instruction-tuned LLMs on factual QA, math reasoning, code synthesis, and creative writing, using six decoders and three seeds. For these models, decoding is not a negligible nuisance variable in any verifiable domain. Untruncated sampling costs 4–7 points on TRIVIAQA, GSM8K, and HUMANEVAL, and top-p or min-p halves that cost. No sampler beats greedy on average. Under an LLM-judge proxy, creative quality does not improve with sampling. Model rankings are reassuringly stable ($\tau \geq 0.85$ vs. greedy in every verifiable domain). Reasoning is the least stable verifiable domain, but it is far from the hypothesized $\tau < 0.7$. Code is the most seed-sensitive domain.

Key takeaway: rankings survive a change of decoder, but individual scores do not. Decoding settings should therefore always be reported and held fixed when comparing models.

Several follow-ups would sharpen these conclusions. The first is to repeat the grid with frontier and unquantized models. The second is to replace the single LLM judge with human ratings or a panel of judges. The third is to add more models (at least 20) for tighter τ estimates. Larger code benchmarks such as MBPP+ [Liu et al., 2023] would help separate inherent code noise from small-n effects, and temperature sweeps would show where each domain's loss begins.

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A Additional Details

Prompts. The user messages, wrapped in each model's chat template, are:

- *Factual:* "Answer the following trivia question. Reply with only the answer (a few words), no explanation. Question: {q} Answer:"
- *Reasoning:* "Solve the following math word problem. Think step by step, then write the final answer on the last line in the form 'The answer is N'. {q}"
- *Code:* "Complete the following Python function. Return the complete function (signature and body, plus any imports it needs) in a single python code block. {prompt}"
- *Creative:* "Write a short, original story of about 150 words that begins with the sentence below. Make it vivid, surprising and engaging. Output only the story. First sentence: {s}"

Per-model ranges of the change relative to greedy. The table below reports the minimum and maximum Δ across the 11 models (in points).

Domain	T=0.7	T=1.0	top-p 0.9	top-k 40	min-p 0.1
Factual	[−4.0, +0.3]	[−8.1, −0.5]	[−7.1, −0.2]	[−6.8, −0.6]	[−3.0, 0.0]
Reasoning	[−6.0, +0.8]	[−14.3, −0.5]	[−8.2, +0.5]	[−13.8, −0.1]	[−8.1, +0.7]
Code	[−7.9, +1.2]	[−15.0, +0.6]	[−7.9, +1.2]	[−15.2, +0.2]	[−6.5, +0.8]
Creative	[−0.09, +0.18]	[−2.15, −0.06]	[−0.24, +0.23]	[−0.18, +0.19]	[−0.07, +0.21]

Table 8: Range of per-model changes relative to greedy. Under T=1.0, every model loses accuracy on factual QA and reasoning.

Models. Checkpoints: `Qwen/Qwen2.5-0.5B-Instruct`, `Qwen/Qwen2.5-1.5B-Instruct`, `Qwen/Qwen2.5-3B-Instruct`, `Qwen/Qwen2.5-7B-Instruct-AWQ`, `Qwen/Qwen3-4B-Instruct-2507`, `hugging-quants/Meta-Llama-3.1-8B-Instruct-AWQ-INT4`, `hugging-quants/gemma-2-9b-it-AWQ-INT4`, `solidrust/Mistral-7B-Instruct-v0.3-AWQ`, `tiiuae/Falcon3-7B-Instruct-AWQ`, `kaitchup/Phi-3-mini-4k-instruct-gptq-4bit`, `kaitchup/Olmo-3-7B-Instruct-gptq-w4a16-g128`. The judge is `Qwen3-8B-AWQ`. We served all models with vLLM 0.32.

Deviations from the plan. We had planned to use a 14B judge, but it did not fit on the shared GPUs, so we used Qwen3-8B-AWQ instead. Because of GPU contention, we adapted the memory budget for each model. Per-request seeds keep outputs independent of these choices. We added the matched-n seed-variance check after seeing the difference in benchmark sizes.